

# **CRYPTOCURRENCY HISTORICAL DATA ANALYSIS USING PYTHON**

## **Project Report**

### **Submitted by:**

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### **Tools Used:**

Python, Google Colab, NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn

## 1. ABSTRACT

This project focuses on the analysis of historical cryptocurrency market data using Python. The dataset contains historical information about multiple cryptocurrencies, including Bitcoin and Ethereum, along with their opening price, closing price, highest price, lowest price, trading volume, and market capitalization. The project applies data preprocessing, statistical analysis, data visualization, correlation analysis, and linear regression techniques. NumPy and Pandas are used for numerical computation and data manipulation, while Matplotlib and Seaborn are used for visualization. SciPy and Scikit-learn are used for statistical and regression analysis. The analysis provides insights into cryptocurrency price trends, daily returns, volatility, and the relationship between Bitcoin and Ethereum.

## 2. INTRODUCTION

Cryptocurrency is a digital form of currency that uses cryptographic techniques to secure transactions and control the creation of new units. The cryptocurrency market contains a large amount of historical price and trading data that can be analyzed to understand market behaviour.

In this project, historical cryptocurrency data is analyzed using Python. The analysis focuses mainly on Bitcoin and Ethereum to study their closing prices, daily returns, trading volume, volatility, and relationship. Various Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn are used for data processing, statistical analysis, visualization, correlation analysis, and linear regression.

The project demonstrates how data science techniques can be applied to real-world financial data to extract meaningful statistical insights from historical cryptocurrency records.

### **3. OBJECTIVES**

The main objectives of this project are:

1. To analyze historical cryptocurrency market data using Python.
2. To perform data cleaning and preprocessing using Pandas.
3. To calculate Bitcoin daily closing price percentage changes using NumPy.
4. To study Bitcoin and Ethereum closing price statistics such as mean and median.
5. To analyze Bitcoin daily return volatility using standard deviation.
6. To visualize cryptocurrency price trends and trading-volume distributions.
7. To study the relationship between Bitcoin and Ethereum using Pearson correlation.
8. To develop a linear regression model to analyze the relationship between Bitcoin and Ethereum prices.
9. To present the results using clear and meaningful data visualizations.

## 4. DATASET DESCRIPTION

The dataset used in this project contains historical market data for multiple cryptocurrencies. Each record represents the market information of a cryptocurrency on a particular date.

The dataset contains **28,944 records and 8 columns**. It includes information for **12 different cryptocurrencies**. The main cryptocurrencies analyzed in this project are Bitcoin and Ethereum.

### Dataset Attributes

| COLUMN     | DESCRIPTION                         |
|------------|-------------------------------------|
| currency   | Name of the cryptocurrency          |
| date       | Date of the recorded market data    |
| open       | Opening price of the cryptocurrency |
| High       | Highest price record                |
| low        | Lowest price recorded               |
| close      | Closing price of the cryptocurrency |
| volume     | Daily trading volume                |
| Market map | Market capitalization               |

The dataset covers historical cryptocurrency records up to **December 4, 2019**. Before analysis, the date column was converted into a proper date format and numerical columns were cleaned and converted into numerical data types.

## 5. TOOLS AND TECHNOLOGIES

### 5.1 Programming Language

**Python** was used as the primary programming language for data preprocessing, numerical calculations, statistical analysis, visualization, and machine learning.

### 5.2 Development Environment

**Google Colab** was used as the development environment to write and execute Python code in a cloud-based Jupyter Notebook environment.

### 5.3 Python Libraries

- **NumPy** – Used for numerical calculations and daily percentage change calculations.
- **Pandas** – Used for data loading, cleaning, filtering, sorting, and manipulation.
- **Matplotlib** – Used to create line graphs, histograms, and scatter plots.
- **Seaborn** – Used for statistical data visualization.
- **SciPy** – Used for statistical analysis and generating the normal distribution curve.
- **Scikit-learn** – Used to build the linear regression model.

### 5.4 Dataset

The project uses a historical cryptocurrency dataset containing price, trading volume, and market capitalization information for multiple cryptocurrencies.

## 6. DATA PREPROCESSING

Data preprocessing is an important step before performing any analysis. The cryptocurrency dataset was first loaded into Python using the Pandas library.

The **Date** column was converted into a proper datetime format so that date-based filtering and sorting could be performed.

The numerical columns such as **Open, High, Low, Close, Volume, and Market Cap** initially contained some values in text format with comma separators. These values were cleaned by removing the commas and converting the columns into numerical data types.

The dataset was then checked for missing values and duplicate records. The cleaned data was sorted by date where required and filtered to obtain individual cryptocurrency datasets such as Bitcoin and Ethereum.

After preprocessing, the cleaned dataset was used for statistical analysis, visualization, correlation analysis, and linear regression.

### 6.1 Data Preparation Steps

1. Load the CSV dataset using Pandas.
2. Convert the **Date** column into datetime format.
3. Remove comma separators from numerical values.
4. Convert numerical columns into appropriate numeric data types.
5. Check for missing values.
6. Check for duplicate records.
7. Filter Bitcoin and Ethereum data for further analysis.
8. Sort the data according to date.

## **7. DATA ANALYSIS AND METHODOLOGY**

After preprocessing the cryptocurrency dataset, different data analysis and statistical techniques were applied to obtain meaningful insights. The analysis was mainly focused on Bitcoin and Ethereum.

### **7.1 Bitcoin Daily Percentage Change**

NumPy was used to calculate the daily percentage change in Bitcoin's closing price. This helps to understand the day-to-day movement of Bitcoin prices.

### **7.2 Ethereum Six-Month Analysis**

Pandas was used to filter Ethereum records for the last six months available in the dataset. This provides a focused view of Ethereum's recent historical data within the dataset.

### **7.3 Bitcoin Closing Price Trend**

A line chart was created using Matplotlib to visualize Bitcoin's daily closing price over time. This helps identify changes and trends in Bitcoin's historical closing prices.

### **7.4 Trading Volume Distribution**

A histogram was used to study the frequency distribution of daily trading volume. This provides an overview of how trading volume values are distributed across the dataset.

### **7.5 Mean and Median Analysis**

The mean and median closing prices of Bitcoin and Ethereum were calculated to understand their central tendency.

### **7.6 Bitcoin Volatility**

Bitcoin daily returns were calculated and their standard deviation was used as a measure of volatility. A higher standard deviation indicates greater variation in daily returns.

## 7.7 Normal Distribution Analysis

A histogram of Bitcoin daily returns was plotted along with a normal distribution curve to examine the distribution of daily price returns.

## 7.8 Bitcoin–Ethereum Correlation

Pearson correlation was calculated to measure the linear relationship between Bitcoin and Ethereum closing prices.

## 7.9 Linear Regression

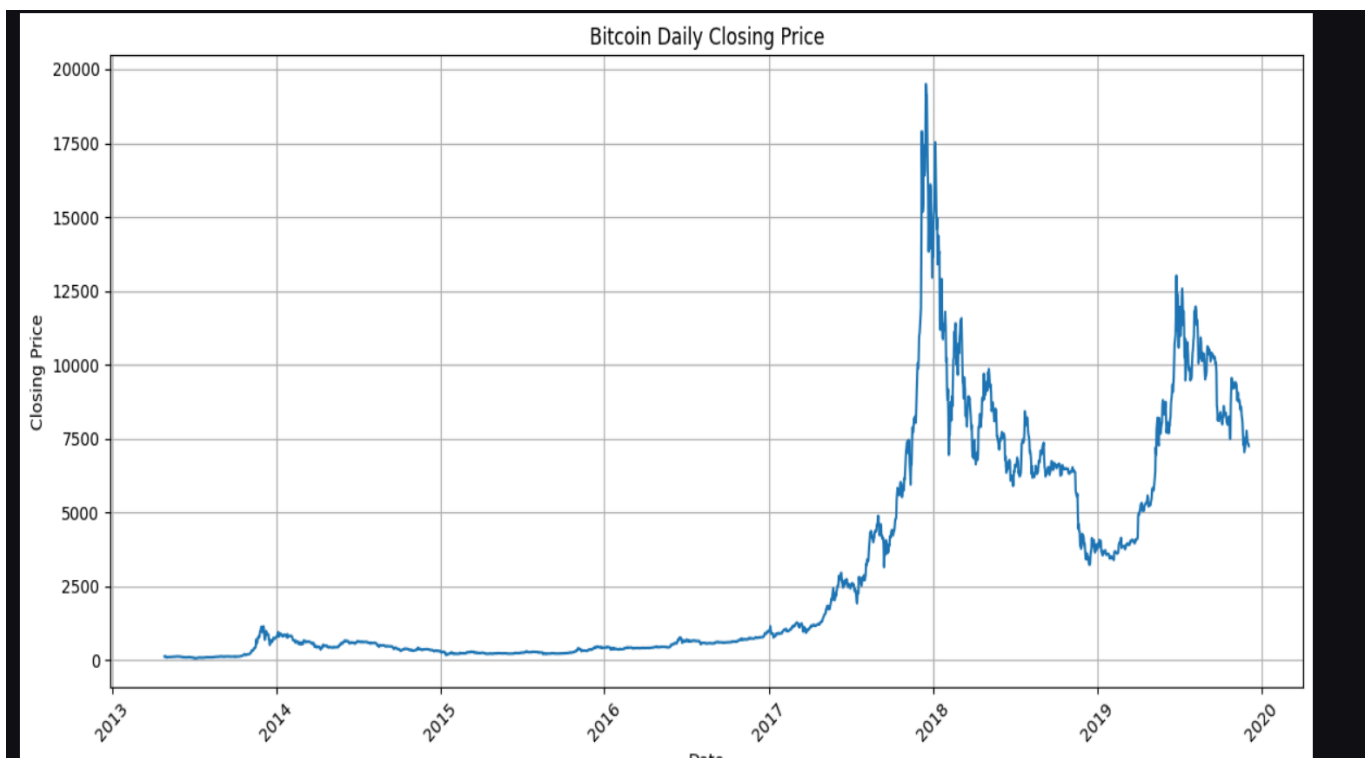
A linear regression model was developed using Bitcoin closing price as the independent variable and Ethereum closing price as the dependent variable. The model was used to describe the linear relationship between the two cryptocurrencies.

# 8. RESULTS AND VISUALIZATIONS

The analysis produced several statistical results and visualizations that helped in understanding the historical behaviour of Bitcoin and Ethereum.

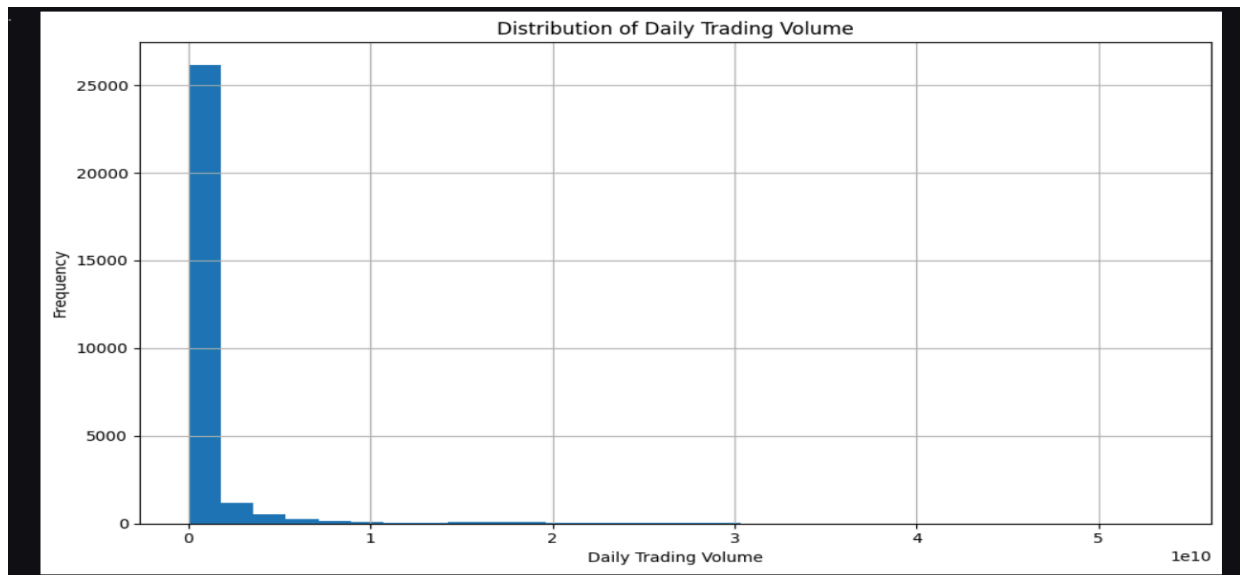
## 8.1 Bitcoin Closing Price Trend

The Bitcoin closing price was visualized using a line chart. The chart shows the changes in Bitcoin's closing price across the available historical period.



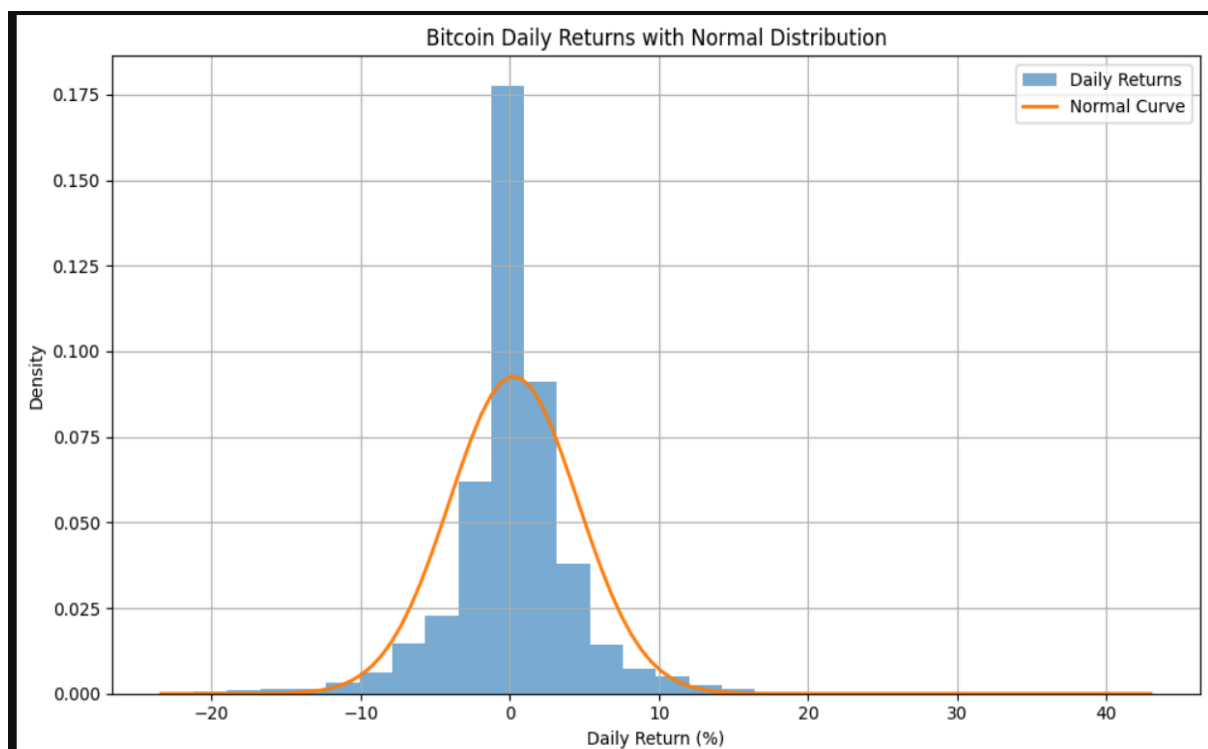
## 8.2 Trading Volume Distribution

A histogram was used to visualize the distribution of daily trading volume. The visualization helps to understand the frequency of different trading-volume values.



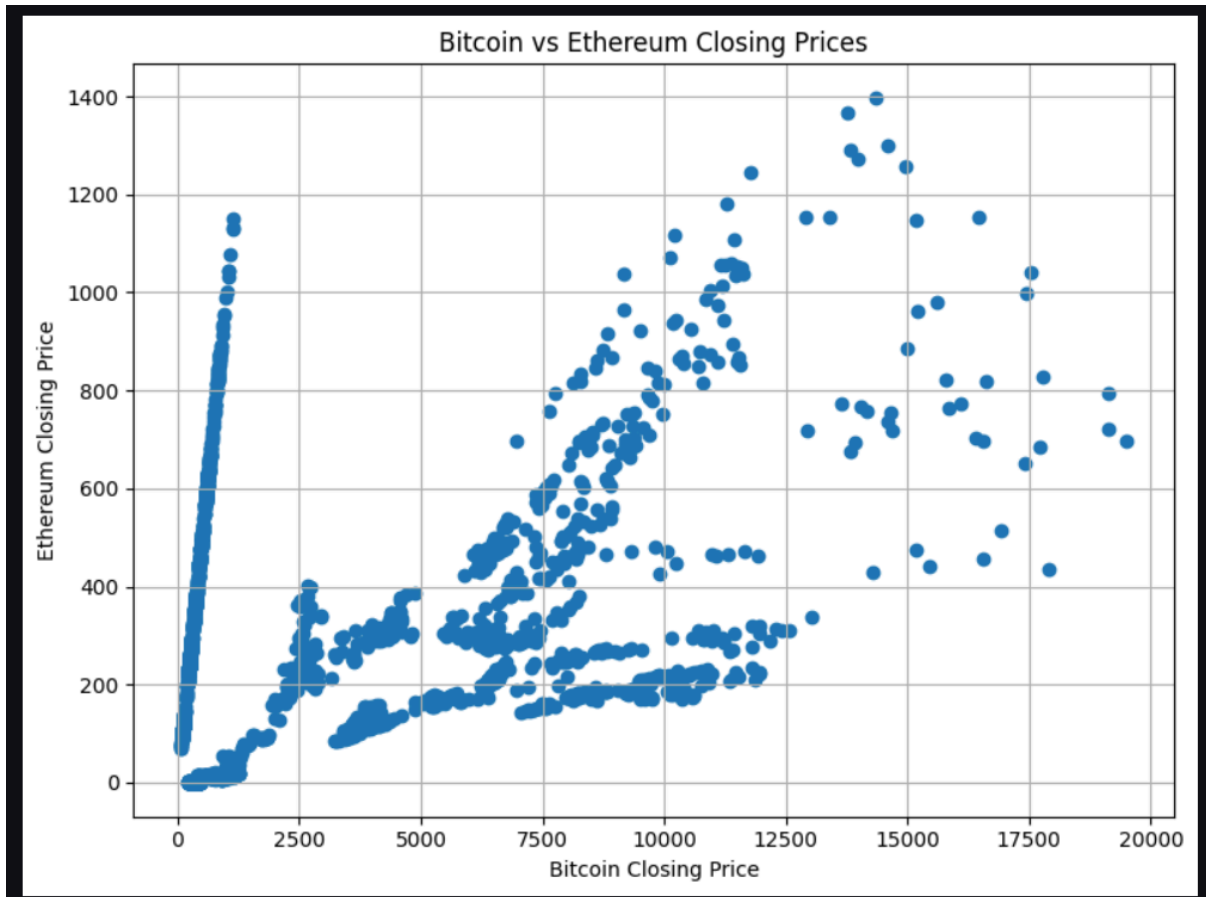
## 8.3 Bitcoin Daily Returns

The daily percentage changes of Bitcoin were analyzed to understand its short-term price movements. A histogram with a normal distribution curve was also used to visualize the distribution of daily returns.



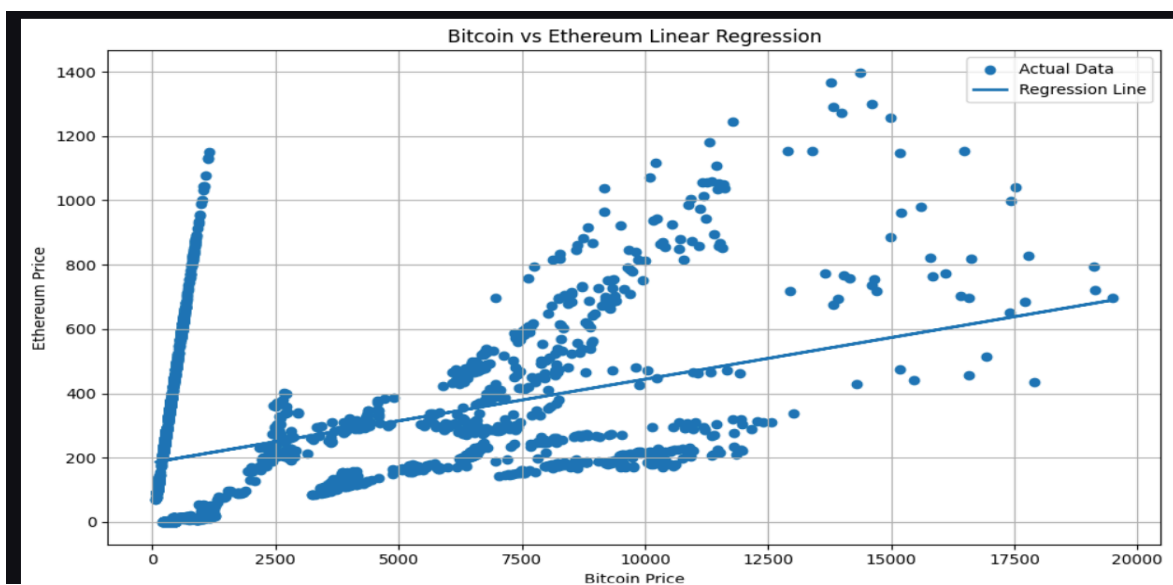
## 8.4 Bitcoin vs Ethereum

A scatter plot was created to examine the relationship between Bitcoin and Ethereum closing prices. Pearson correlation was then calculated to quantify their linear relationship.



## 8.5 Linear Regression

Linear regression was applied to describe the relationship between Bitcoin and Ethereum closing prices. The regression line was plotted together with the actual observations



## 9. STATISTICAL RESULTS

The statistical analysis of Bitcoin and Ethereum provided the following results.

| Metric               | Bitcoin  | Ethereum |
|----------------------|----------|----------|
| Mean Closing Price   | 3023.499 | 263.435  |
| Median Closing Price | 701.945  | 211.345  |

### Bitcoin Return Analysis

| Metric               | Result |
|----------------------|--------|
| Average Daily Return | 0.258% |
| Standard Deviation   | 4.303% |

### Correlation and Regression

| Analysis             | Result   |
|----------------------|----------|
| Pearson Correlation  | 0.3883   |
| Regression Slope     | 0.0259   |
| Regression Intercept | 185.1195 |

### Regression Equation:

$$Ethereum = 185.1195 + 0.0259 \times Bitcoin$$

## 10. KEY FINDINGS

The major findings obtained from the cryptocurrency historical data analysis are as follows:

1. Bitcoin had a mean closing price of **3023.499**, while its median closing price was **701.945**.
2. Ethereum had a mean closing price of **263.435**, while its median closing price was **211.345**.

3. The average daily return of Bitcoin was approximately **0.258%**.
4. The standard deviation of Bitcoin daily returns was **4.303%**, indicating variation in its daily price movements.
5. The Pearson correlation between Bitcoin and Ethereum closing prices was **0.3883**, indicating a positive linear relationship in the analyzed dataset.
6. The linear regression model produced a slope of **0.0259** and an intercept of **185.1195**.
7. Data visualizations such as line charts, histograms, scatter plots, and regression plots helped in understanding cryptocurrency price trends, return distribution, and relationships.

## 11. CONCLUSION

This project analyzed historical cryptocurrency market data using Python and various data science techniques. The dataset was cleaned and processed using Pandas, while NumPy was used for numerical calculations.

The analysis focused mainly on Bitcoin and Ethereum. Statistical measures such as mean, median, average daily return, and standard deviation were calculated to understand cryptocurrency price behaviour and volatility. Matplotlib and other visualization techniques were used to identify price trends and trading-volume distributions.

The Pearson correlation analysis showed a positive linear relationship between Bitcoin and Ethereum closing prices in the analyzed dataset. Linear regression was also applied to describe this relationship mathematically.

Overall, this project demonstrates how Python-based data analysis can be used to process historical cryptocurrency data, visualize market behaviour, and obtain meaningful statistical insights. The results describe patterns in the historical dataset and should not be considered as predictions of future cryptocurrency prices.

## 12. FUTURE SCOPE

The project can be further improved by using larger and more recent cryptocurrency datasets. Future analysis can include a greater number of cryptocurrencies and a longer historical period.

Advanced machine learning and time-series forecasting techniques can also be applied to study cryptocurrency price movements. Technical indicators such as moving averages, RSI, and MACD can be included for deeper analysis.

An interactive dashboard can be developed using tools such as Streamlit or Power BI to present cryptocurrency trends and statistical results in an interactive manner.

Future work can also evaluate prediction models using proper training and testing datasets and performance metrics.